Programming Psychology:

A Digital Laboratory for the Algorithms of Human Cognition

<u>Psychophysics Demo</u>

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1. Abstract (The Elevator Pitch)

“Programming Psychology" is an interactive digital “laboratory” designed to transform abstract psychological theories into tangible, user-generated data. Traditional psychology education often relies on static definitions to describe dynamic cognitive processes, creating a pedagogical gap between memorization and experience. Currently focused on the foundational domain of Psychophysics, the laboratory functions as a suite of computational instruments calibrated to measure the Difference Threshold (“Just Noticeable Difference”). Users engage with precise sensory trials, manipulating variables like depth, luminance, and color-value in real-time to locate the exact point where a stimulus change becomes noticeable. By validating established laws, such as Weber’s Law, through active experimentation rather than passive observation, the project aims to establish a self-measurable quantitative baseline for human sensation. This foundation lays the necessary groundwork for future perceptual organization modules (Gestalt), Memory, and Decision-Making, ultimately aiming to generate a unique "perceptual signature" for each user.

2. Introduction

The Pedagogical Gap: Memorization vs. Application

This project emerged from noticing a stark contrast in pedagogy of Interactive Multimedia (IMM) and Psychology. Design and programming are inherently active disciplines; one cannot prove they understand "visual hierarchy" or "loops" without building something.

Psychology education, conversely, often relies heavily on passive memorization of text-based definitions. Concepts that are dynamic and experiential, such as sensory processing or perceptual organization, are reduced to static terms on a page. I realized that while I could *define* these concepts for a standardized test, I lacked a tool to apply or experience my learning. As a response to that gap, I aim to apply the active, "learning-by-doing" methodology of creative coding to the theoretical landscape of psychology.

Methodology: The Interactive Textbook

Drawing inspiration from the "sketching with code" philosophy found in *Generative Design (Gross, Bohnacker, Hartmut, Julia, Lazzeroni)* and the interactive learning models of p5play (Quinton Ashley, Paolo Pedercini), this project reimagines the psychology textbook as an interactive, computational environment. In design, we test ideas by iterating on prototypes; "Programming Psychology" applies this same methodology to cognitive science. Rather than reading about Weber's Law (the principle that the Just Noticeable Difference is proportional to the magnitude of the stimulus), users manipulate the stimulus themselves. The learner moves from a passive recipient of information into an active participant, allowing users to test psychological theories through personally-generated data rather than static memorization.

Clinical Aspirations: Quantifying the Subjective

Beyond its educational value, this project is foundational to my future career in Clinical Psychology. Empathy is often distinguished from sympathy by the ability to truly understand another's lived reality. Just as nurses studying gerontology (the study of aging) wear weighted suits to physically understand the mobility limitations of the elderly, or designers use simulators to experience color blindness, I believe technology can help us temporarily "inhabit" the perceptual reality of others.

While clinical tools already exist to quantify sensory thresholds, they often function as "black boxes," accessible only to experts and obscuring the logic behind the score. "Programming Perception" prioritizes data accessibility. By visualizing the user's performance metrics in real-time using clear, readable interfaces rather than obscure tabular data, the project demystifies the psychophysical process. It ensures that the result of the experiment is just as understandable as the experience of it.

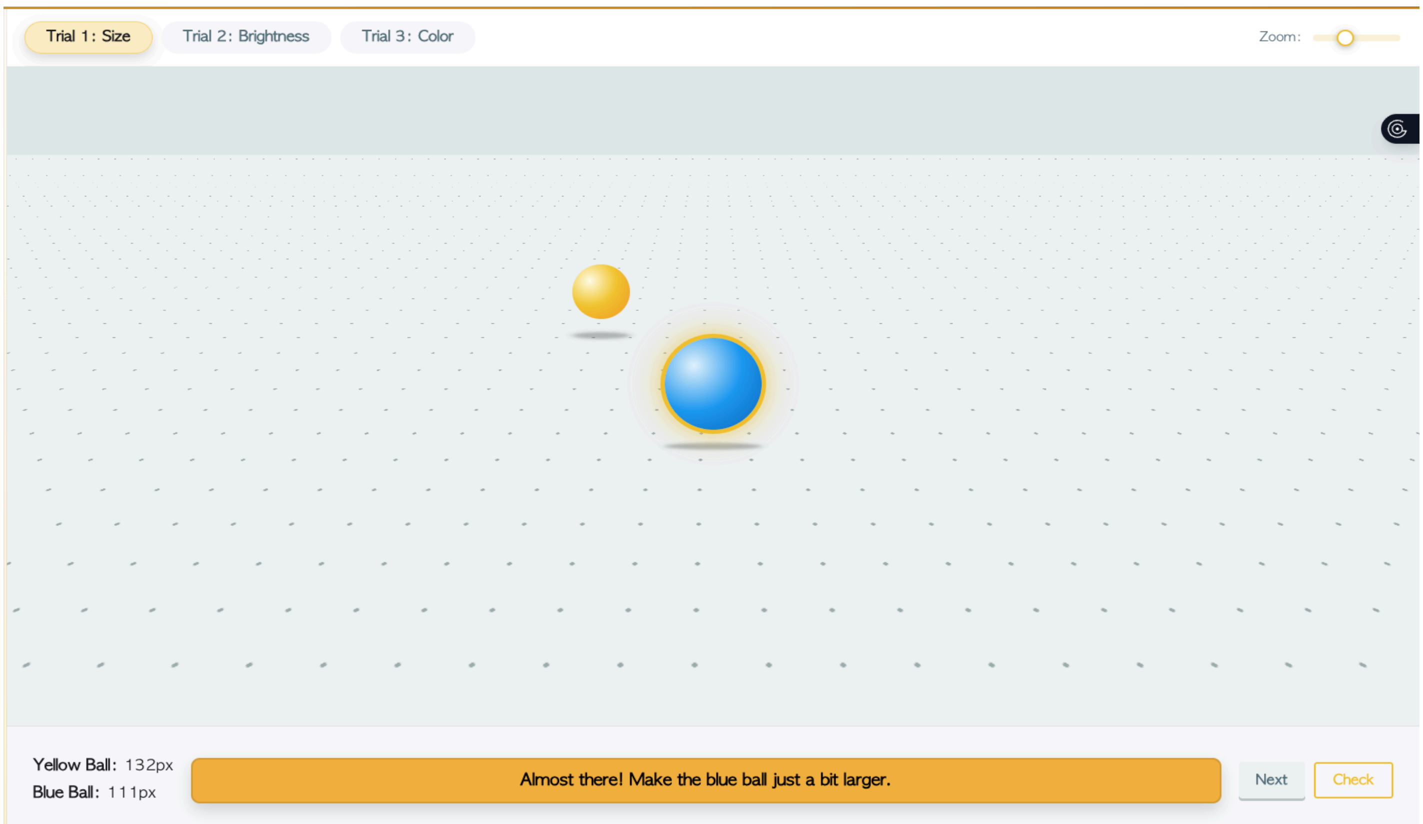
3. Historical Context: Quantifying the Invisible

The Convergence of Psychophysics and Generative Art

This project operates at the intersection of two historical lineages that sought to quantify subjective human experience: Psychophysics and Generative Art. In the 19th century, Gustav Fechner founded Psychophysics with the goal of proving the mind could be measured with the same rigor as the physical world. He figured out that sensation follows mathematical rules, such as Weber's Law (the Just Noticeable Difference) and the Absolute Threshold.

A century later, computer art pioneers like Vera Molnár had a similar goal in the fine art field. Molnár used early computers not to create "pictures," but to systematically investigate the "Machine Imaginaire, "using algorithms to explore the mathematical rules of visual form.

The Psychophysics section of "Programming Psychology" merges these traditions. It applies the computational methods of generative art, specifically algorithmic variation of form, to answer the scientific questions posed by Fechner. Both fields attempt to quantify abstract concepts into tangible results for; Fechner quantified sensation, Molnár quantified aesthetics, and this project uses the tools of the latter to visualize the former.



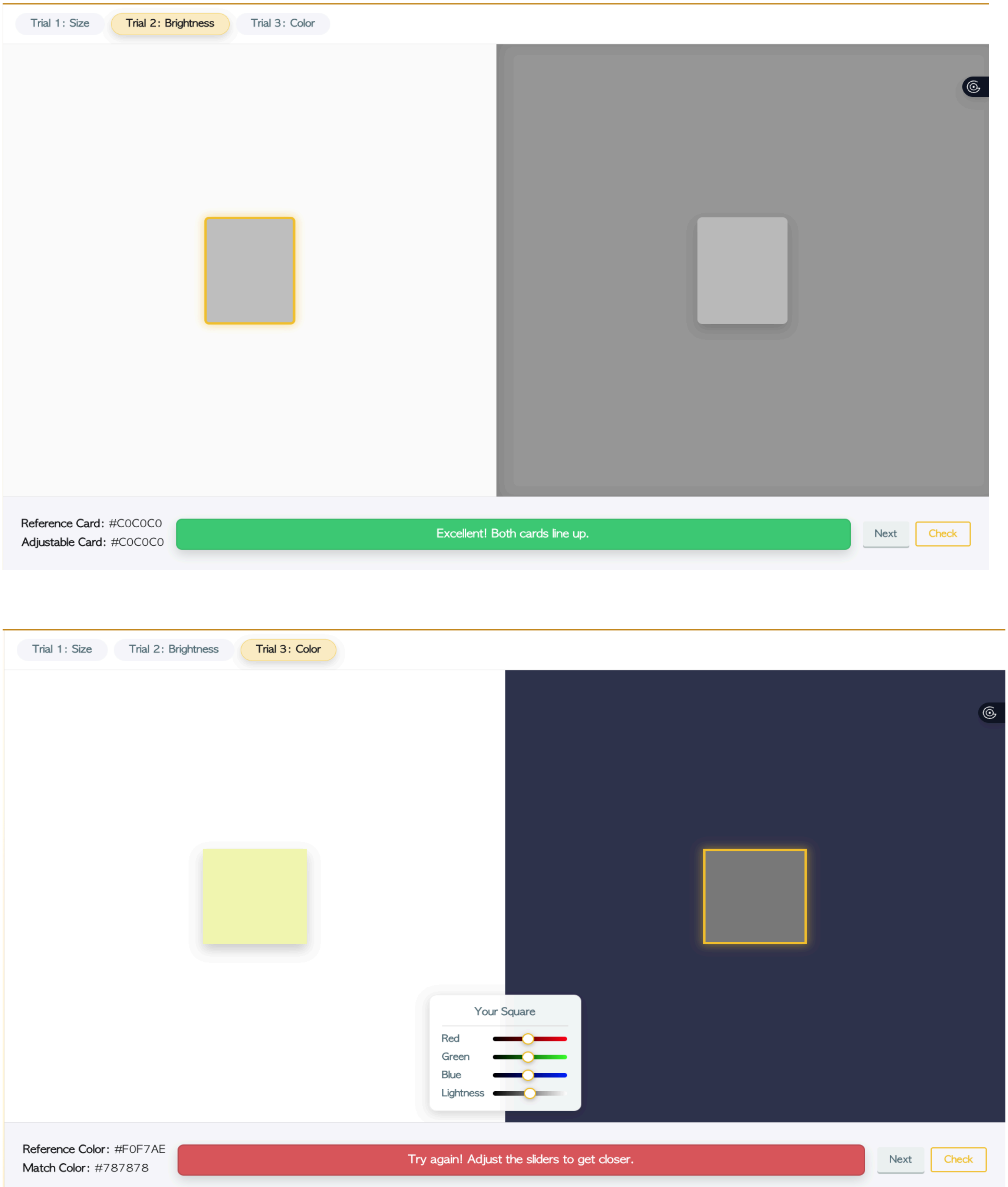


Fig. 1: Size, brightness, and color difference threshold modules.

The Evolution of the Instrument: Democratizing the Lab

Historically, the study of psychophysics was confined to elite laboratories, requiring heavy brass instruments or expensive specialized computers. This project moves the lab into the web browser, marking the next stage in that technological evolution. The project democratizes access to scientific equipment by reconstructing these traditional experiments using common web technologies (HTML/CSS/JavaScript). Before ever entering a formal research setting, this accessibility turns the user from a passive reader into an active scientist who can push their senses.

Methodology: The Aesthetic of Constraint

The project currently adopts a minimalist, grid-based visual language. While this aligns with modern interface design standards and trends, it also serves a critical scientific function described in computational theory as the "aesthetic of constraint". In a psychophysics experiment, any extraneous visual information acts as a confounding variable.

By reducing the interface to simple geometric forms on a controlled grid, the system minimizes visual "noise," ensuring that the user’s interaction is driven purely by their sensory threshold. I allow simple gradients within my UI and I will be able to examine if even those simple additions are confounding over usability testing.

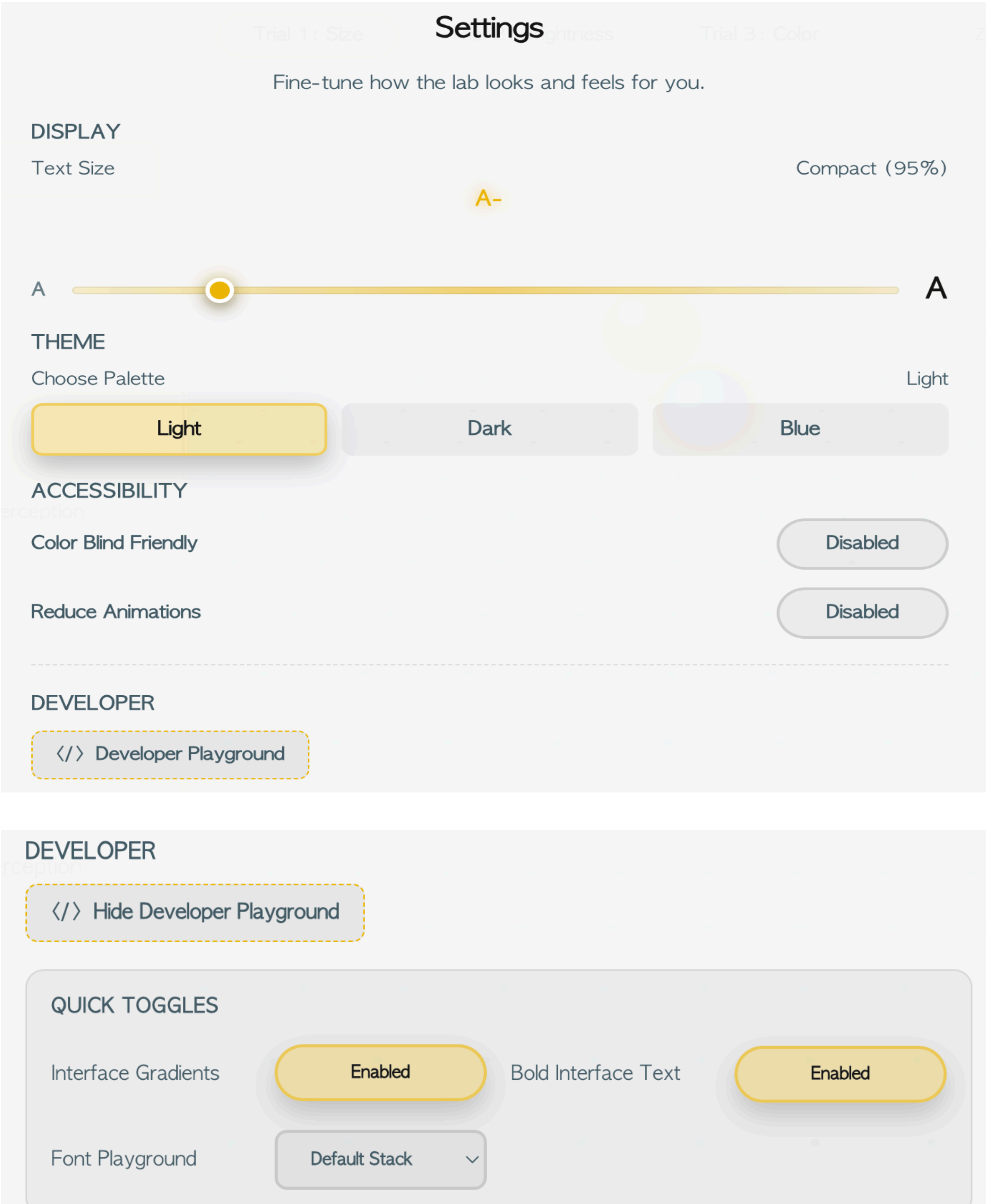


Fig. 2: User-controlled accessibility settings, utilized to ensure the interface accommodates different visual needs without compromising the experimental data.

4. Current Context: Bridging Play and Rigor

The Gap Between "Dry Data" and "Creative Play"

The current landscape of digital tools for psychology and creative coding remains largely divided. On one side, standard psychological learning resources (such as online GRE study guides or academic lab software) focus heavily on "data" and "memorization." While scientifically rigorous, they often lack engagement, presenting concepts like Weber's Law as static formulas or definitions for a graded test.

On the other side, creative coding educators like Daniel Shiffman (The Coding Train) and Patt Vira use code for "play" and "aesthetics," teaching dynamic interaction but rarely applying it to rigorous scientific measurement.

"Programming Psychology" occupies the middle ground between these two. It applies the engaging, "sketching with code" techniques of the creative coding community, such as real-time feedback loops and responsive variables, to build a clinical instrument. It effectively gamifies the calibration of the human eye, using play as a way to branch into formal measurements and testing.

Differentiation: Generative Feedback vs. Static Animation

Currently, a search for "Weber's Law" yields an AI-overview and static images of the equations ($\Delta I / I = k$). This forces students to memorize a formula without understanding its physical reality. Other existing online resources, such as gestaltprinciples.com, serve as valuable references but often rely on pre-made motion graphics or simple "click-to-reveal" demonstrations. These are passive experiences as the user watches an animation play out.

My generative approach is fundamentally different because the output is based on the user’s specific input. The code does not just "show" a threshold, it returns feedback from the user’s unique actions. This transforms the screen from a display monitor into a responsive environment, examining the user's perception through immediate data from pre-determined tests. Embodying the core pedagogical philosophy of Generative Design, the user moves from the "performer" role, drawing an image, to a "conductor" role, orchestrating a dialogue between the digital system (the code) and the biological system (the eye).

The Pedagogical Value of "Modern" Design

Ultimately the project adopts a contemporary app design style (gentle gradients, dark/light themes and sleek UI elements). In a time when well-made apps are the standard, a "worn-out" or old-fashioned academic interface can instantly discourage engagement. By presenting tasks through a user-friendly visual approach the tool conveys that the information is relevant, up-to-date and deserving of attention.

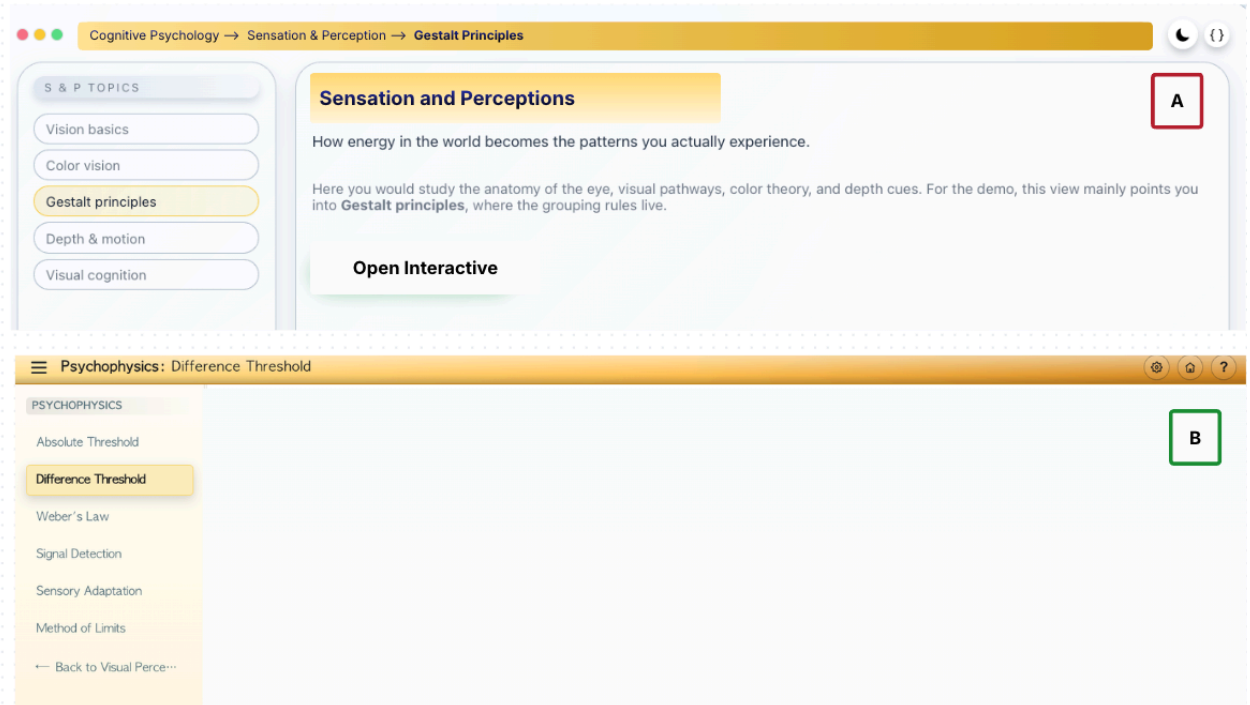


Fig. 3: UI development showing Iteration 1 (A), the early Gestalt Principles interface, and Iteration 2 (B), the refined Psychophysics Difference Threshold layout.

5. Technical Strategy: Precision, Modularity, and Constraint

The Precision of the DOM: Vanilla JavaScript & HTML5

During Phase 1 (Sensation & Perception) the app employs Vanilla JavaScript along with native HTML5 input elements. Although canvas libraries offer capabilities they frequently add abstraction layers that may conceal raw data. By using Document Object Model (DOM) components, like <input type="range"> the app takes advantage of the browsers inherent accessibility and ensures accurate numerical outputs (for instance 0-255, for RGB color channels). This architectural choice ensures that the "scientific instrument," the slider itself, remains stable and predictable.

```css
.color-col label { min-width: 70px; }
.color-col input[type=range] { width: 120px; }
```

Fig. 4: CSS Slider Width Settings for the Color-Matching Module.

Technical Necessity of 3D:

While 2D DOM manipulation is sufficient for testing brightness or color thresholds, it is inadequate for rigorously testing depth perception. Current 2D implementations relying on simple perspective illusions (arbitrarily scaling an object as it moves "up" the Y-axis) lack the mathematical fidelity required for a true psychophysics trial. To ensure scientific validity, particularly for testing monocular cues like linear perspective and motion parallax, a true three-dimensional coordinate system is essential. Therefore, the "Depth & Motion" modules will utilize p5.js WebGL mode within an instance wrapper. This introduces a mathematically accurate Z-axis, ensuring that changes in perceived object size are derived from actual geometric distance calculations rather than illustrative 2D tricks, thereby making the trial psychophysically "usable" and accurate.

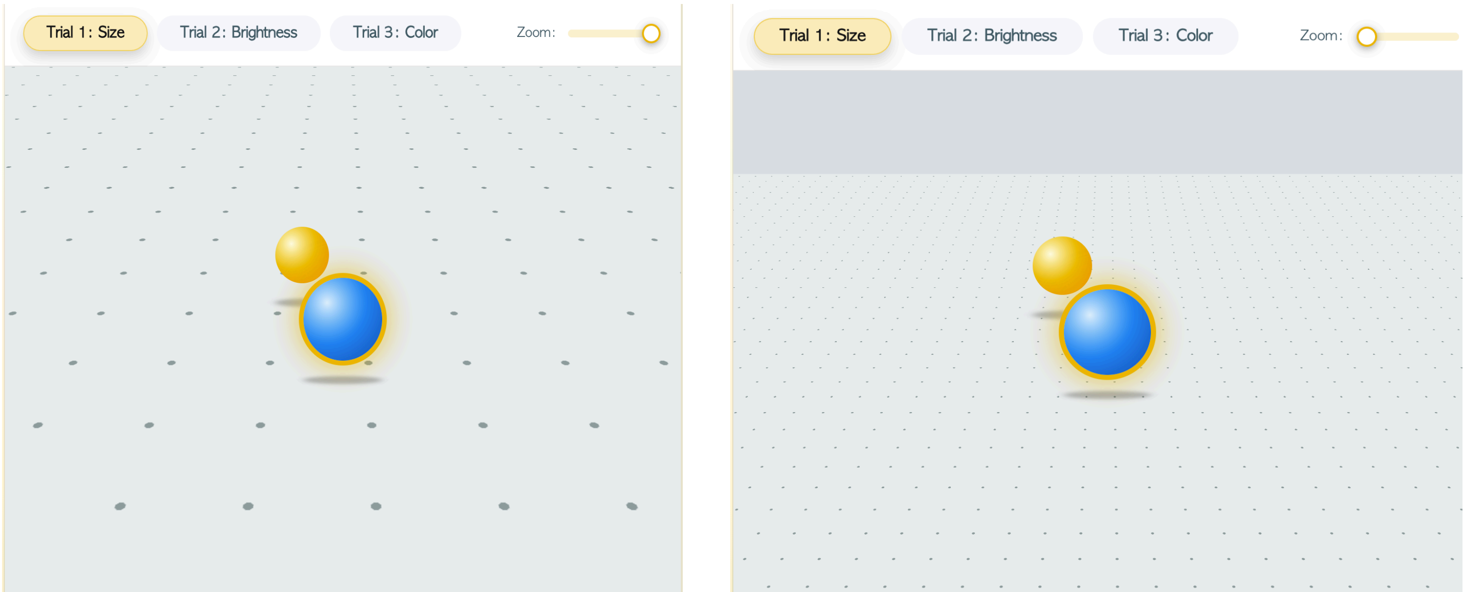


Fig. 5: Limits of UI for depth and size tasks without using p5.js webGL, showing the plane being zoomed out does not change the position of the 2D circles.

Data Persistence: The Local Lab

To transition the tool from a fleeting "game" to a functional "lab," the prototype implements **LocalStorage** (client-side browser storage). This provides an elegant, lightweight solution for data persistence without the immediate complexity of a backend database. It allows users to refresh the page and instantly see their "Lowest Threshold" or historical performance, creating a personal feedback loop.

In future iterations, this local data architecture will serve as the foundation for a scalable cloud-based login system, enabling longitudinal tracking of a user’s "perceptual signature."

Methodology: Sketching with Variables

Finally, the codebase is designed to support the "sketching with code" methodology advocated by *Generative Design*. Key perceptual variables such as the depthScale used in the Size

Discrimination trial (currently initialized at 0.5), are exposed as tunable parameters. This structure allows for iterative discovery; these "magic numbers" are not fixed constants but flexible values that can be fine-tuned based on user testing to ensure the perspective illusions feel accurate and "fair."

```
if (currentLevel === 'size') {
    state.size.refSize = Math.floor(Math.random() * 80) + 80;
    state.size.target = state.size.refSize;
    state.size.depthScale = 0.5;
    const visualSize = state.size.refSize * state.size.depthScale;
```

Fig. 6: CSS Slider Width Settings for the Color-Matching Module.

6. Implementation: The Cognitive Foundation for Expansion

The implementation strategy is designed to prove the scalability of the "Programming Psychology" framework.

<table>
<tr><td colspan="3">Table 1: The Cognitive Core (Dec – Feb)
Goal: Develop a fully functional, depth-first interactive suite for Cognitive Psychology. This phase builds the reusable "Code Skeletons" (e.g., game-container, data-logger, instruction-modal) that will power the rest of the application.</td></tr>
<tr><th>Sub-Domain</th><th>Key Psychological Concepts</th><th>Technical Implementation</th></tr>
<tr><td rowspan="3">A. Sensation & Perception
(Vision, Hearing, Psychophysics)</td><td>• Thresholds: Complete Difference/Absolute Threshold trials.</td><td rowspan="3">Slider Engine: A logic block for measuring user inputs against a target value.</td></tr>
<tr><td>• Gestalt: Implement p5.js sketches for Proximity & Closure.</td></tr>
<tr><td>• Multimodal: Add WebAudio API for pitch detection (Hearing).</td></tr>
</table>

B. Memory Models	• **Iconic Memory:** A "flash" test measuring visual retention (ms).	**Timer Engine:** Precise millisecond tracking for reaction time and retention duration.
	• **Working Memory:** A "Visuospatial Sketchpad" pattern-matching game.	
	• **Long-Term:** A "priming" experiment using semantic associations.	
C. Thinking & Decision-Making	• **Problem Solving:** "Tower of Hanoi" / "Candle Problem" demo.	**Logic Engine:** State-management systems that track multi-step user problem-solving paths.
	• **Heuristics:** A sorting task to demonstrate Availability Heuristic or Confirmation Bias.	

Table 2: The Structural Stress-Test (March – April)

Goal: Use the assets from Phase 1 to rapidly prototype **one fully complete demonstration** for each of the remaining 5 domains. This proves the project is a scalable "Textbook Supplement" rather than just a toy.

Sprint	Domain	Representative Prototype Ideas
1	**Biological Psychology**	**The Neuron Builder:** Use the "Slider Engine" to adjust ion concentrations (Na+/K+) and trigger an Action Potential animation.
2	**Methods & Measurements**	**The Correlation Sandbox:** An interactive scatterplot where users drag data points to see how r-value (correlation coefficient) changes in real-time.
3	**Developmental Psychology**	**The Conservation Task:** A digital replication of Piaget's water task. Users pour digital liquid into different shaped containers to test their own conservation logic.

4	**Social Psychology**	**The Conformity Test:** A simulation of the Asch Line Experiment where "bot" users give wrong answers to pressure the real user.
5	**Clinical Psychology**	**The Anxiety Baseline:** A modified Psychophysics trial measuring reaction time to negative vs. neutral words (Stroop Effect variant) to visualize emotional bias.

Table 3: Synthesis & Launch (Late April) **Goal:** Final integration and usability testing
The "Home" System: Finalize the main navigation dashboard that houses all 6 categories (Cognitive, Bio, Methods, Dev, Social, Clinical) in the "Retro-OS" file system.
Data Export: Ensure the localStorage system correctly saves a "Perceptual Signature" that includes data points from all completed modules.
Documentation: Compile the version history and work on promotional material. Highlighting the **reuse of code for ease of creation** (e.g, showing how the same slider used for *Weber's Law* (Phase 1) could be repurposed for another task later.

7. Summary and Conclusion:

The Synthesizer: Design, Code, and Psychology

"Programming Psychology" is more than a technical demonstration; it is a proof of concept for a new kind of interdisciplinary rigor. It demonstrates that I can design with the aesthetic sensitivity of an artist, architect software with the logical precision of a developer, and investigate the human mind with the seriousness of a psychologist.

My goal is to explore the complexity of the human experience, and I have found that the most effective way to teach these complexities is to build systems that allow others to inhabit them.

This project is the combining of my academic journey, applying the research methods of psychology to the generative potential of code.

Educational Impact: From Passive to Active

If integrated into a high school or undergraduate curriculum, this tool has the potential to transform the classroom dynamic. Instead of students passively absorbing definitions of Weber’s Law while fighting off sleep, they could be actively debating whose color vision is more acute or whose depth perception is more precise. By functioning as a dynamic companion to traditional homework, the application turns abstract theory into personal competition and discovery. It provides a visual language for students who, like myself, struggle to connect with static text, ensuring that the study of perception is as vivid as perception itself.

Future Trajectories and Final Vision

As I prepare for graduate study in Clinical Psychology, this project serves as a scalable framework. The architecture developed here, specifically the Generative design "Conductor" model of user interaction, can be expanded to visualize more abstract clinical concepts, such as anxiety thresholds or cognitive distortions. Ultimately, this thesis is built on a singular, profound realization: both computer code and the human brain run on **invisible algorithms**. One dictates the logic of the machine; the other dictates the logic of our reality. "Programming Psychology" will reveal both.

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